paradox conveniently disappears because what is good for one gene is good for all.

But what if a gene exerted a phenotypic effect that was good for itself but bad for the rest of the genes in the body? This is not a flight of fancy. Cases of it are known, for instance the intriguing phenomenon called meiotic drive. Meiosis, you will remember, is the special kind of cell division that halves the number of chromosomes and gives rise to sperm cells or egg cells. Normal meiosis is a completely fair lottery. Of each pair of alleles, only one can be the lucky one that enters any given sperm or egg. But it is equally likely to be either one of the pair, and if you average over lots of sperms (or eggs) it turns out that half of them contain one allele, half the other. Meiosis is fair, like tossing a penny. But, though we proverbially think of tossing a penny as random, even that is a physical process influenced by a multitude of circumstances—the wind, precisely how hard the penny is flicked, and so on. Meiosis, too, is a physical process, and it can be influenced by genes. What if a mutant gene arose that just happened to have an effect, not upon something obvious like eye colour or curliness of hair, but upon meiosis itself? Suppose it happened to bias meiosis in such a way that it, the mutant gene itself, was more likely than its allelic partner to end up in the egg. There are such genes and they are called segregation distorters. They have a diabolical simplicity. When a segregation distorter arises by mutation, it will spread inexorably through the population at the expense of its allele. It is this that is known as meiotic drive. It will happen even if the effects on bodily welfare, and on the welfare of all the other genes in the body, are disastrous.

Throughout this book we have been alert to the possibility of individual organisms ‘cheating’ in subtle ways against their social companions. Here we are talking about single genes cheating against the other genes with which they share a body. The geneticist James Crow has called them ‘genes that beat the system’. One of the best-known segregation distorters is the so-called *t* gene in mice. When a mouse has two *t* genes it either dies young or is sterile. *t* is therefore said to be ‘lethal’ in the homozygous state. If a male mouse has only one *t* gene it will be a normal, healthy mouse except in one remarkable respect. If you examine such a male’s sperms you will find that up to 95 per cent of them contain the *t* gene, only 5 per cent the normal allele. This is obviously a gross distortion of the 50 per cent ratio that we expect.